A Frame-Semantic Deep Learning Approach for Suicide Risk Detection on Social Media

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Agenda

1. Introduction & Motivation
2. Problem Statement
3. Methodology & Proposed Framework
4. Experimental Setup
5. Results & Analysis
6. Conclusion & Future Work

1. Introduction & Motivation

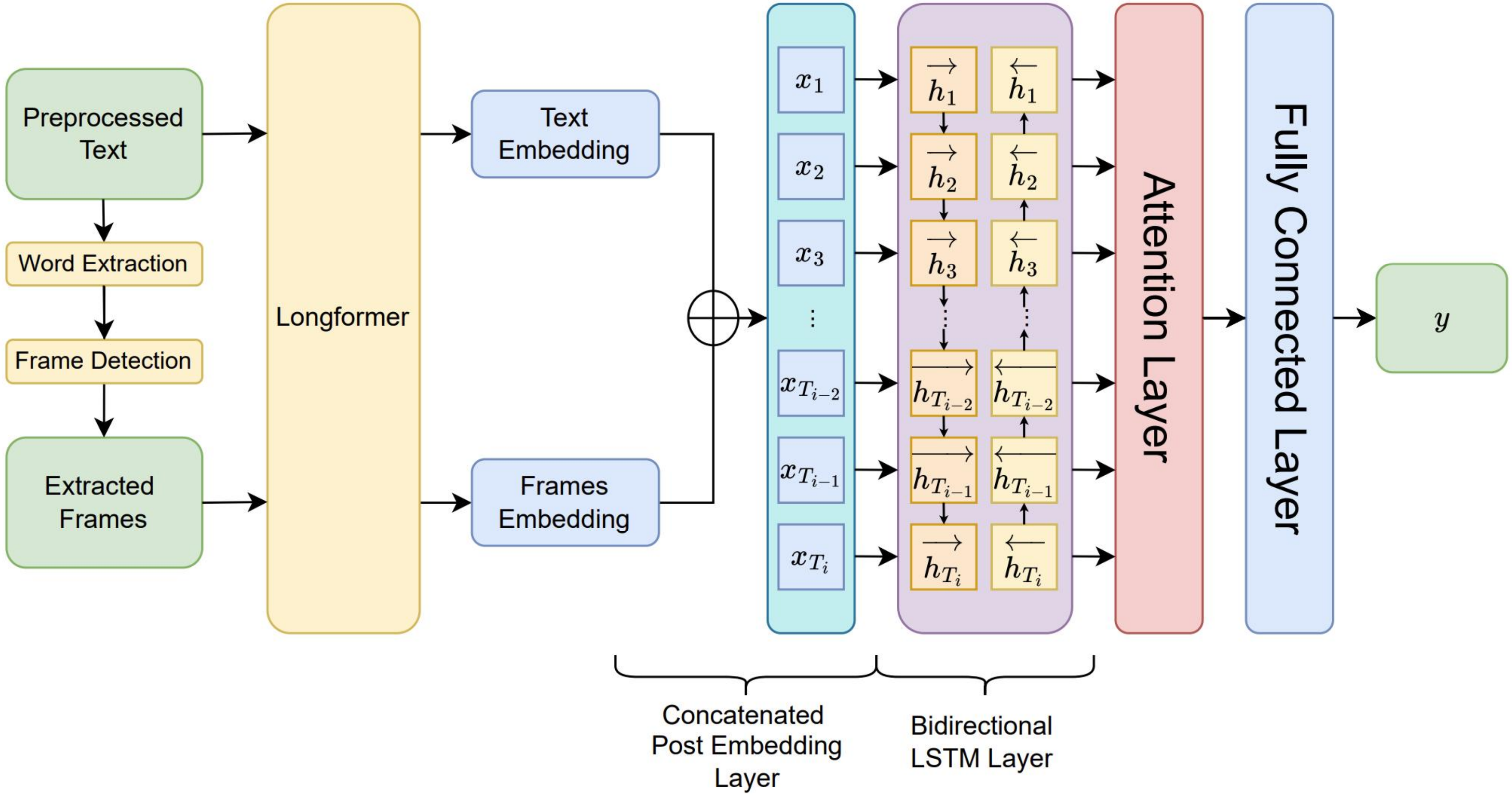
- **The Problem**: Suicide is a significant global health crisis, responsible for over 700,000 deaths annually (WHO).
- **The Opportunity**: Many individuals express distress on social media, providing a crucial window for early detection and intervention
- **The Gap**: Automated models struggle to interpret the implicit and metaphorical language often used to express suicidal thoughts.
- **Inspiration:** Cognitive Metaphor Theory (CMT)

2. Problem Statement

Given a user's social media posts, classify their suicide risk level.

Given user $u_i \in U = \{u_1, u_2, \ldots, u_n\}$, our objective is to evaluate their suicide risk level $y \in Y$. Here, $Y = \{l_1, l_2, \ldots, l_k\}$ is a pre-defined set of k ordinal risk categories, representing increasing levels of severity. Each user u_i has a sequence of posts $P_i = \{p_i^1, p_i^2, \ldots, p_i^{T_i}\}$ over time, where p_i^t denotes the user's t^{th} post, with $p_i^{T_i}$ is the most recent post and T_i is the user's total number of posts.

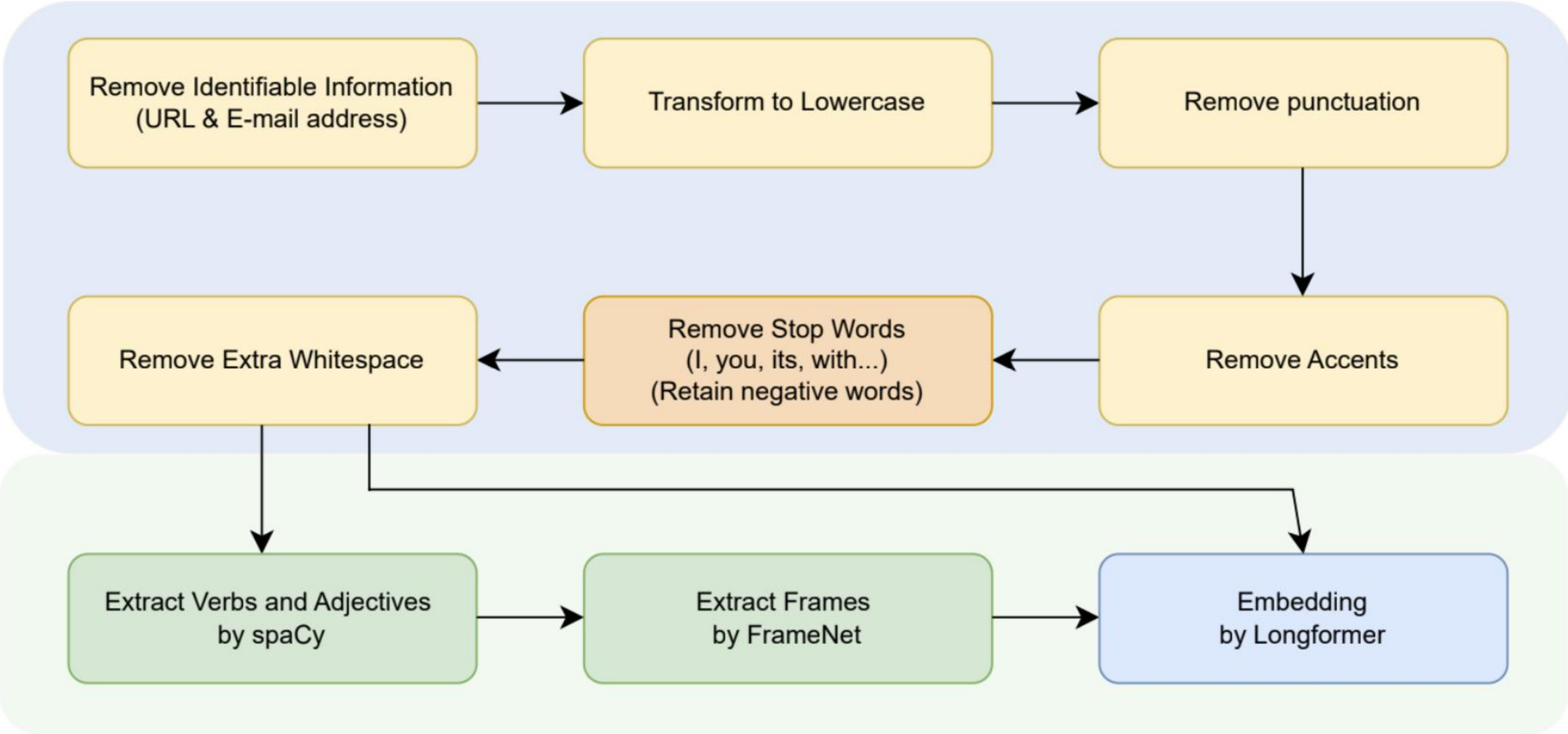
3. Methodology & Proposed Framework



4. Experimental Setup

Risk Level (Class-wise Distribution)	Definition
Supportive [SU] (20%)	Users who engage in discussion without using language indicative of being at-risk in the past or the present.
Suicidal Indicator [IN] (20%)	Users who use at-risk language but are not actively experiencing any general or acute symptoms.
Suicidal Ideation [ID] (34%)	Users who express thoughts of suicide or show pre-occupations with known risk factors.
Suicidal Behavior [BR] (15%)	The confession of active or historical self-harm.
Actual Attempt [AT] (9%)	The confession of any deliberate action that could have resulted in intentional death, be it a completed attempt or not.

4. Experimental Setup



4. Experimental Setup

Baselines:

- XGBoost with TF-IDF features
- Longformer
- Bi-LSTM+Attention (without frames extraction)

Evaluation Metrics:

- F-Score
- Ordinal Error (OE)

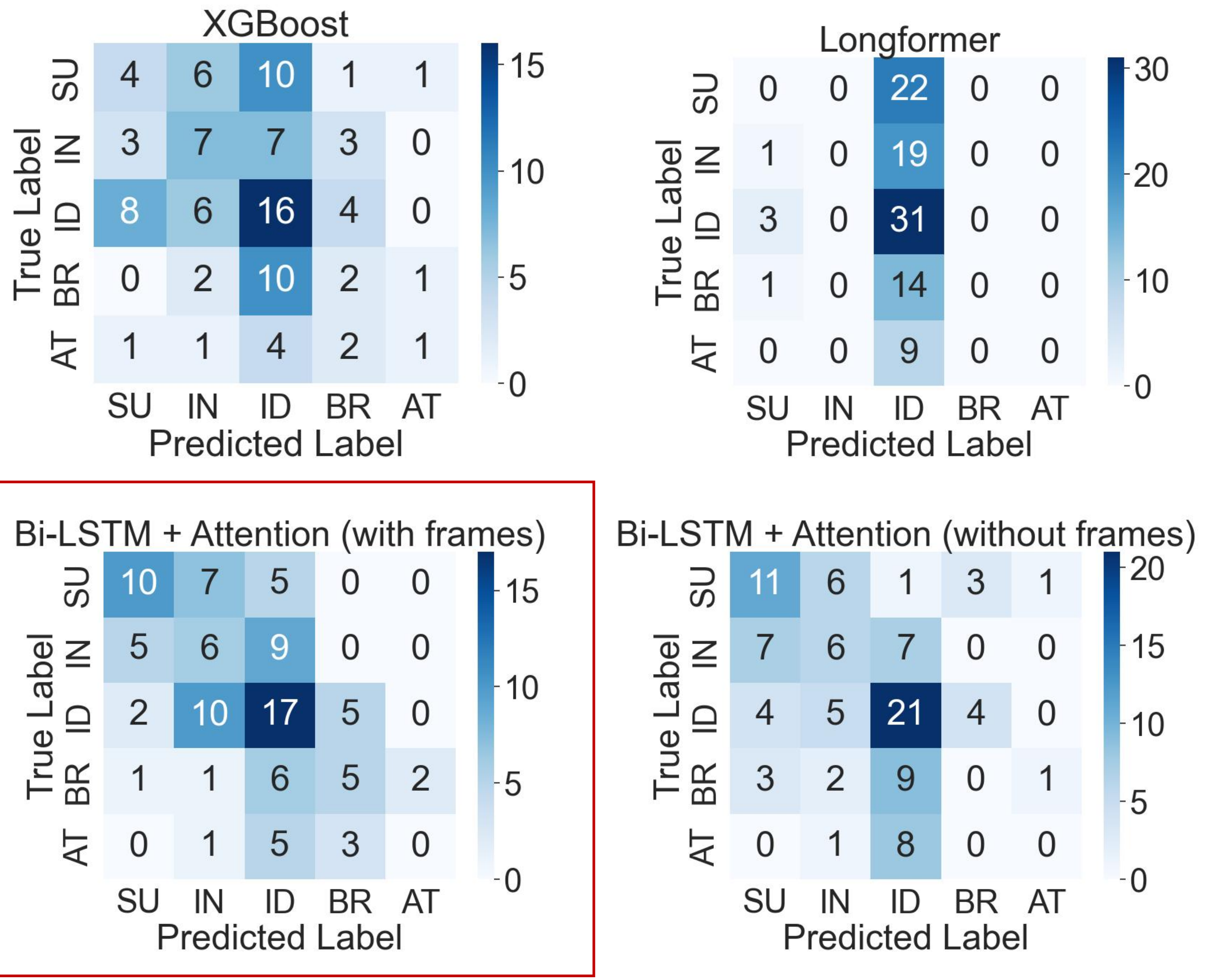
$$FP = \frac{\sum_{i=1}^{N_T} I(r_i^p > r_i^a)}{N_T}, \quad FN = \frac{\sum_{i=1}^{N_T} I(r_i^a > r_i^p)}{N_T}, \quad OE = \frac{\sum_{i=1}^{N_T} I(|r_i^a - r_i^p| > 1)}{N_T}$$

5. Results & Analysis

Table 2: Experiment with 5-label Classification

Model	Input	Graded Precision	Graded Recall	F-Score	OE
XGBoost	TF-IDF vector	0.48	0.45	0.46	0.31
Longformer	Text+Frames	0.43	**0.53**	0.47	0.35
Bi-LSTM+Attention	Text+Frames	0.58	**0.53**	**0.55**	**0.15**
Bi-LSTM+Attention	Text	**0.62**	0.49	**0.55**	0.23

5. Results & Analysis



5. Results & Analysis

Table 3: Frames extraction from a case study

Text	True Label	Extracted Frames	Predicted Label
"I just want to die somewhere … She gets angry at me, destroys me, breaks me … Tired rarely she gives me some love … I'm Tired of this world."	Attempt	Emotion_directed Personal_relationship Desirability	Ideation Ideation $\boxed{\textbf{Behavior*}}$ Ideation

Note: The predicted labels correspond to the following models, in order: XGBoost, Longformer, our proposed model (with frames), and the ablation model (without frames). The asterisk (*) denotes the best prediction.

6. Conclusion & Future Work

Summary of Work:

- Proposed and validated a novel framework

Ethical Consideration:

- Sensitive topic
- Tool as assistance, not for replace

Future Work:

- A benchmark dataset (fully evaluated by professors)
- Multi-task learning
- Combined with Large Language Models

References

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